

5G RAN

NR Inter-Carrier Dynamic Power Allocation Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 03 (2025-08-30).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationships of NR inter-carrier dynamic power sharing and NR multi-band power adaptive scheduling with RF module fast deep dormancy. For details, see 4.3.2.3 Function Impacts and 5.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Editorial Changes

Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-061222	NR Inter-Carrier Dynamic Power Sharing	4 NR Inter-Carrier Dynamic Power Sharing
FOFD-090100	Multi-band Power Adaptive Scheduling	5 NR Multi-Band Power Adaptive Scheduling (Low-Frequency TDD)

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
NR inter-carrier dynamic power sharing	None	4 NR Inter-Carrier Dynamic Power Sharing
NR multi-band power adaptive scheduling (low-frequency TDD)	None	5 NR Multi-Band Power Adaptive Scheduling (Low-Frequency TDD)

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
NR inter-carrier dynamic power sharing	Supported only in low frequency bands	4 NR Inter-Carrier Dynamic Power Sharing
NR multi-band power adaptive scheduling (low-frequency TDD)	Supported only in low frequency bands	5 NR Multi-Band Power Adaptive Scheduling (Low-Frequency TDD)

3 Overview

At a given moment, the bandwidth of NR carriers can be fully occupied or become idle. Power allocation between multiple NR carriers allows a carrier with fully occupied bandwidth to use the power of idle carriers if further data scheduling is required. In this way, the instantaneous power available to this carrier may exceed its statically configured power.

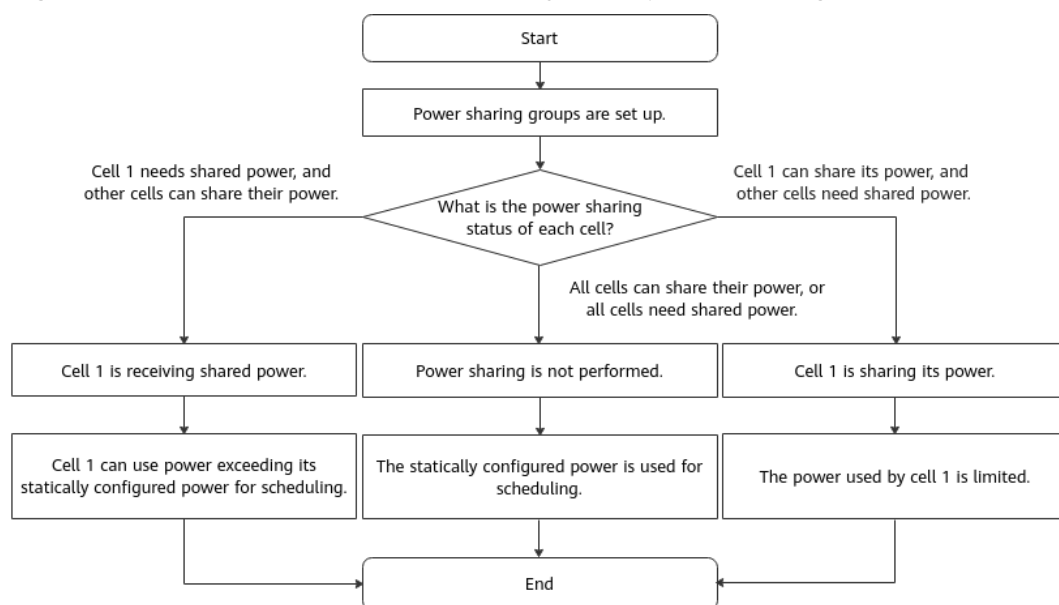
- NR inter-carrier dynamic power sharing
Power is shared between multiple NR carriers on the same RF channel.
- NR multi-band power adaptive scheduling (low-frequency TDD)
Power adaptive scheduling is performed between multiple NR carriers on the same RF module.

4 NR Inter-Carrier Dynamic Power Sharing

4.1 Principles

NR inter-carrier dynamic power sharing enables TTI-level power sharing. This function is controlled by the **NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch.DynPowerSharingSwitch** parameter. [Figure 4-1](#) shows the procedure of this function.

Figure 4-1 Procedure of NR inter-carrier dynamic power sharing



Procedure of NR Inter-Carrier Dynamic Power Sharing (TTI-Level)

1. Power sharing group setup

After this function is enabled, the gNodeB automatically sets up power sharing groups for NR cells on the RF channels when all of the following conditions are met:

- The cells have the same number of downlink transmit channels.

- The cells share power amplifiers (PAs).
- For NR TDD, the cells have the same slot configuration.
- For NR TDD, the cells have the same subcarrier spacing.
- There are no carriers with a bandwidth less than 5 MHz on the RF channels of the module.

Each BBP can accommodate a maximum of 12 power sharing groups.

For TDD, each power sharing group supports power sharing between two cells.

In **Figure 4-2**, four channels (corresponding to PA 1 to PA 4) are used as an example.

- Cell 1 and cell 3 can form a power sharing group because they share PAs and have the same number of downlink transmit channels.
- Cell 2 and cell 3 cannot form a power sharing group because they have different numbers of downlink transmit channels.
- Cell 2 and cell 4 cannot form a power sharing group because not all of their PAs are shared.

Figure 4-2 Example of power sharing group setup

	RF channel corresponding to PA 1	RF channel corresponding to PA 2	RF channel corresponding to PA 3	RF channel corresponding to PA 4
Cell 1				
Cell 2				
Cell 3				
Cell 4				

2. Evaluation of the power sharing status of each cell

The gNodeB checks the power sharing status of each cell in a power sharing group based on the volume of the data scheduled in the previous TTI. It evaluates whether a cell is sharing power, receiving shared power, or not participating in power sharing.

3. Calculation of the power used by a cell

NR inter-carrier dynamic power sharing cannot increase the RF channel capability, but it can increase user-perceived throughput by reallocating power from idle cells to busy cells. When this function is enabled, the sum of power used by all cells in a power sharing group cannot exceed the sum of their static power configurations.

- For a cell receiving shared power:

Power used by the cell in a TTI = Statically configured power + Shared power received

Maximum shared power received by the cell in a TTI = Statically configured power of the cell x Maximum percentage of shared power

 NOTE

- "Statically configured power" refers to the power manually configured for a cell.
- For a cell sharing its power: Power used by the cell in a TTI = Statically configured power – Power shared
- For a cell not participating in power sharing: Power used by the cell in a TTI = Statically configured power

4.2 Network Analysis

4.2.1 Benefits

It is recommended that this function be enabled when traffic is instantaneously imbalanced between NR cells that share RF channels in light-load scenarios. This increases the **User Downlink Average Throughput (DU)** excluding the last slot (Mbit/s). For counters related to gain measurement, see [4.4.3 Network Monitoring](#).

 NOTE

$$UserDownlinkAverageThroughput(DU) = \frac{\sum N.ThpVol.DL.cell - \sum N.ThpVol.DL.LastSlot.cell}{\sum N.ThpTime.DL.RmvLastSlot.cell}$$

User Downlink Average Throughput (DU) increases by:

- 5% to 20% if the percentage of the last TTIs before the downlink buffer is empty is less than 40%
- 2% to 5% if the percentage of the last TTIs before the downlink buffer is empty is greater than or equal to 40%

 NOTE

Percentage of the last TTIs before the downlink buffer is empty (%) = $(N.ThpTime.DL - N.ThpTime.DL.RmvLastSlot) / N.ThpTime.DL \times 100\%$

The preceding benefits are obtained if both of the following conditions are met:

- The average downlink physical resource block (PRB) usage is less than or equal to 40%.
Average downlink PRB usage (%) = $N.PR.B.DL.Used.Avg / N.PR.B.DL.Avail.Avg \times 100\%$
- The average downlink MCS index is less than or equal to 15.

 NOTE

UEs using 256QAM in the downlink do not benefit from NR inter-carrier dynamic power sharing.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- Changes in the application status of this function will trigger parameter adjustments, which cause traffic fluctuations for around 2 minutes and affect the cell throughput. Therefore, it is recommended that this function be enabled when traffic is light.
- This function increases the power consumed by the PDSCH, increasing the average MCS index and reducing the downlink PRB usage.
- The impact of this function on the downlink packet delay (indicated by the **N.QoS.DL.PktDelaygNBDU.Time** counter) is related to services. That is, when the percentage of tail packets (the percentage of the last TTIs before the downlink buffer is empty) is low, the downlink packet delay decreases. Conversely, when the percentage of tail packets is high, the downlink packet delay increases.
- This function increases the RF module power usage. Although a base station consumes more power, its power consumption does not exceed the maximum power consumption of all its cells working under full load.
- This function adjusts the power of the cells in a power sharing group in each TTI, which requires information to be exchanged between the cells. This may slightly increase the CPU load of BBPs (indicated by **VS.BBUBoard.CPULoad.Mean**) to no more than 5%.
- This function evaluates the power sharing status and allocates power at the TTI level based on scheduling results in the previous TTI. The evaluation may be inaccurate as NR services are running in real time. Therefore, the access delay increases. You are not advised to enable this function when testing the uplink and downlink peak data rates on the gNodeB.
- (TDD) Parameter adjustments following changes in the application status of this function change the coverage of common and control messages, such as synchronization signal and PBCH block (SSB), channel state information reference signal (CSI-RS), and physical downlink control channel (PDCCH). Such changes may cause the access success rate, handover success rate, service drop rate, and paging success rate to deteriorate.
- After this function is enabled, data packets transmitted in the last TTIs before the downlink buffer is empty are not involved in power sharing, and data packets involved in power sharing are transmitted faster in the TTIs other than these last TTIs. Therefore, the percentage of such last TTIs will increase.

4.3 Requirements

4.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
TDD	FOFD-061222	NR Inter-Carrier Dynamic Power Sharing	NR0S00CD PS00	gNode B	Per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Multi-position PDCCH scheduling	ANY_OCCASION_PD CCH_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter	None
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCell NetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_POWER_ADAPT_SWITCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>
Low-frequency TDD	Slot-level power control	POWER_CONTROL_SLOT_SW option of the NRDUCellSmartPwrLock.SmartPowerLockSw parameter	None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power aggregation in unoccupied bandwidths	NRDUCellChnPwr.MaxPdschConvPwrOffset	<i>Power Control</i>	When NR inter-carrier dynamic power sharing and power aggregation in unoccupied bandwidths are both enabled, the latter function may provide lower gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power reduction for CCUs	NRDUCellChnPwr.MaxUeSatPwrOffset	<i>Power Control</i>	When NR inter-carrier dynamic power sharing and power reduction for CCUs are both enabled, the gains of power sharing may decrease.
Low-frequency TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INTRF_AVOID_SW option of the NRCCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	When interference coordination between public and dedicated networks is enabled, the gains of NR inter-carrier dynamic power sharing may decrease.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When intra-base-station DL CoMP is enabled, the gains of NR inter-carrier dynamic power sharing may decrease.
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW or INTER_GNB_DL_CS_CBF_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	When coordinated interference management is enabled, the gains of NR inter-carrier dynamic power sharing may decrease.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellColLabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When Coordinated Massive MIMO (Co-MM) is enabled, the gains of NR inter-carrier dynamic power sharing may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When data convergence in slots is enabled for power saving scheduling, a cell will share less power with others, affecting the gains of power sharing.
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both NR inter-carrier dynamic power sharing and energy saving based on flexible frequency-domain scheduling take effect, the latter function provides lower gains. As a result, the gains of power sharing decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	- If the PWR_SHR_IN_POWER_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgorithmSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of timing carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. - If the PWR_SHR_IN_POWER_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgorithmSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	- If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of intelligent carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. - If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgorithmSwitch parameter is deselected for a cell, power sharing is not performed on the cell within the period specified for RF channel intelligent shutdown. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgorithmSwitch parameter is selected for a cell, power sharing is not performed on the cell when the cell is in the energy saving state. When a cell enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: - In intra-base-station multi-carrier scenario s: INTRA_GNB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter and FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter - In inter-base-station multi-carrier scenario s: INTER_GNB_M	<i>Energy Conservation and Emission Reduction</i>	- If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of fast carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. - If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
		ULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingSwitch</i> parameter and FAST_CARRIER_SHUT_DOWN_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingExtSwitch</i> parameter Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingExtSwitch</i> parameter		

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Inter-carrier dynamic power sharing is not performed during RF channel shutdown.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	A carrier exits a power sharing group once it enters the deep dormancy state.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYSW (5G gNodeB, LTE eNodeB) set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	A carrier exits a power sharing group once it enters the super dormancy state.
Low-frequency TDD	Non-Slot symbol-level scheduling for low latency	NON_SLOT_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter	None	The user-perceived rates of Non-Slot UEs may decrease in a cell sharing its power.
Low-frequency TDD	DL 256QAM	NRDUCellAlgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	UEs using 256QAM cannot benefit from NR inter-carrier dynamic power sharing.
Low-frequency TDD	CSI-RS rate matching	CSIRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchingSwitch parameter	<i>Downlink Scheduling</i>	When CSI-RS rate matching takes effect, the gains of NR inter-carrier dynamic power sharing may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	In Hyper Cell networking, the gains of NR inter-carrier dynamic power sharing may decrease.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUT_DOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When NR inter-carrier dynamic power sharing and TTI-level channel shutdown both take effect, the energy saving gains of TTI-level channel shutdown decrease and the downlink UE throughput may decrease.
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	In Fusion Cell networking, NR inter-carrier dynamic power sharing may take effect for a lower proportion of cases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency convergent scheduling	MULTI_FREQ_CONVERGENT_SCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	- Multi-frequency convergent scheduling enables real-time scheduling for individual CA UEs to solve the problem of inaccurate downlink data split between the PCell and SCells. - NR inter-carrier power sharing allocates the estimated remaining power of a carrier to other carriers for a CA UE if the carrier is not allocated enough data as a result of inaccurate downlink data split between the PCell and SCells. Therefore, both multi-frequency convergent scheduling and NR inter-carrier power sharing can improve the downlink experience of CA UEs. When both functions are enabled, the benefits provided by either function may decrease.
Low-frequency TDD	PDCCH aggregation level reduction	PDCCH_AGG_LVL_COMPRESS_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Only NR inter-carrier dynamic power sharing takes effect when it is enabled together with PDCCH aggregation level reduction.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellDMAmc.DLModulationSchemeSw parameter	<i>Modulation Schemes</i>	If there are UEs for which DL 1024QAM is being used, the gains of NR inter-carrier dynamic power sharing decrease.
Low-frequency TDD	Low power consumption mode (NR)	LOW_PWR_MODE_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	When both low power consumption mode (NR) and NR inter-carrier dynamic power sharing are enabled and a customized backup power saving policy is used, only low power consumption mode (NR) takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Maximum SSB power offset	NRDUCellTrpBeam.MaxSsbPwrOffset	<i>Power Control</i>	For low-frequency NR TDD 32T32R or higher RF modules other than AAU5x1x, when the NRDUCellTrpBeam.MaxSsbPwrOffset parameter is set to a value greater than 3 dB and NR inter-carrier dynamic power sharing takes effect: • If the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter is deselected, the SSB power aggregation capability decreases. • If the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter is selected, NR inter-carrier dynamic power sharing does not work in slots with SSB transmission. This prevents the decrease in the SSB power aggregation capability, but this power sharing function no longer delivers gains in these slots.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart PDCCH symbol allocation	PDCCH_SYM_SMART_ALLOC_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	When enabled together with smart PDCCH symbol allocation, NR inter-carrier dynamic power sharing may take effect in a lower proportion of cases, therefore providing lower gains.
Low-frequency TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When NR inter-carrier dynamic power sharing takes effect, CCE partial symbol adaptation provides lower gains.
Low-frequency TDD	Downlink precise RB estimation	PRECISE_DL_RB_ESTIMATE_SW option of the NRDUCellDISch.DlSchAlgoSwitch parameter	<i>Downlink Scheduling</i>	When enabled together with downlink precise RB estimation, NR inter-carrier dynamic power sharing may provide lower gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	A carrier exits a power sharing group once RF module fast deep dormancy takes effect for this carrier. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

For TDD, the main control board must be a UMPT, and the baseband processing unit must be a UBBPfw, UBBPg, or later board. (This function does not work in 1T, 2T, 4T, or 8T cells served by the UBBPfw.)

RF Modules

For TDD, the following RF modules do not support this function:

- RRU5254
- RRU5258 (2600 MHz)

Cells

- For TDD:
- Among cells served by the same RF module, NR inter-carrier dynamic power sharing can be enabled for only two of them in each NR band.
 - If the **NRDUCell.DuplexMode** parameter is set to **1T2R**, **2T2R**, **2T4R**, **4T4R**, or **8T8R**, the **NRDUCell.SsbSubcarrierSpacing** parameter must be set to **30KHZ**.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dynamic Power Sharing Switch	NRDUCellAlgoSwitch.DynPowerSharingSwitch	Select the NR_DYN_POWER_SHARING_SW option.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling NR inter-carrier dynamic power sharing for the cell with a local cell ID of 0
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,DynPowerSharingSwitch=NR_DYN_POWER_SHARING_SW-1;
```

Deactivation Command Examples

```
//Disabling NR inter-carrier dynamic power sharing for the cell whose local cell ID is 0  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,DynPowerSharingSwitch=NR_DYN_POWER_SHARING_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

To verify that this function has been activated, perform the following steps:

- Step 1** Run the **DSP NRDUCELLTRPDYNPWRSHR** command on the gNodeB. If the value of **NR Dyn Power Sharing Valid Status** is **Enabled**, this function has been enabled.
- Step 2** Run the **DSP GNODEBDYNPWRSHRGRP** command on the gNodeB to check whether power sharing groups are successfully set up and which cells are in the same power sharing group.
- Step 3** Observe the counters listed in [Table 4-2](#). If the counter values are not zero, this function has taken effect.

Table 4-2 Counters related to NR inter-carrier dynamic power sharing

Counter ID	Counter Name
1911823028	N.PwrShare.PwrOut.Slot.Num
1911823027	N.PwrShare.PwrIn.Slot.Num

 NOTE

The **N.PwrShare.PwrIn.Slot.Num** counter indicates the actual number of TTIs within which the cell uses received power. As power sharing uses the estimation results from the previous TTI, the actual number of TTIs within which the cell uses the received power might differ from the estimated number.

----End

4.4.3 Network Monitoring

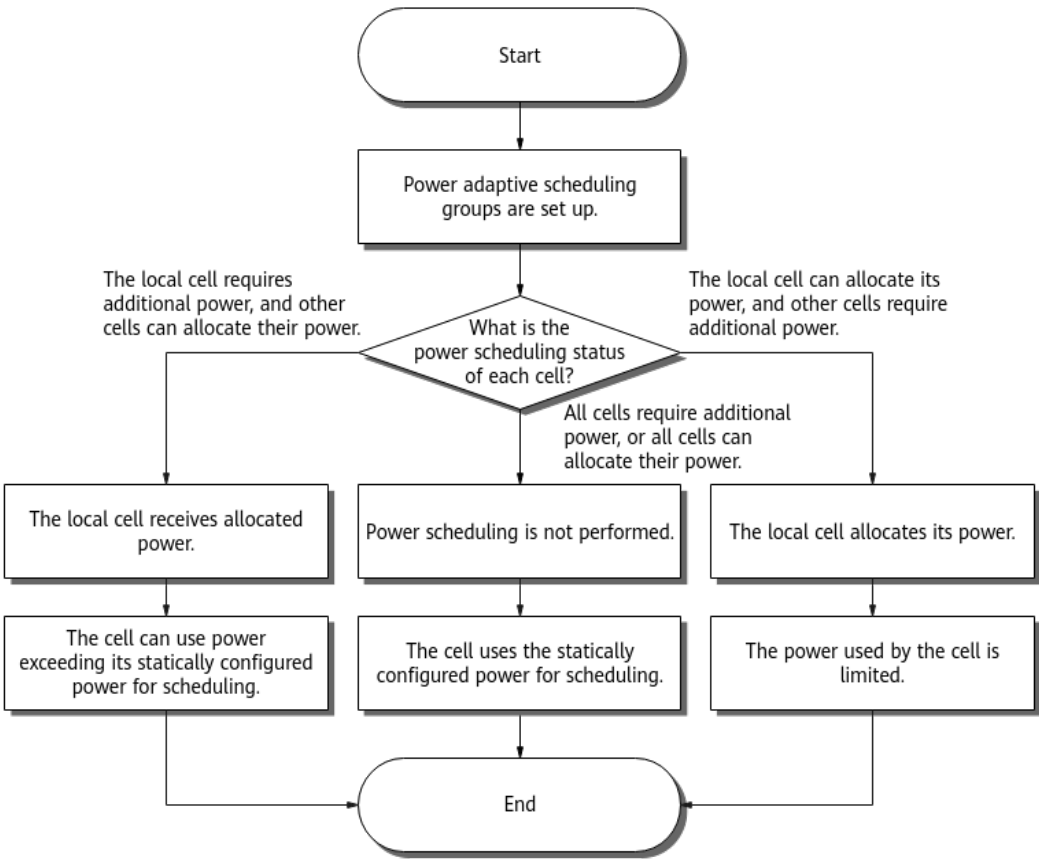
Compare the values of **User Downlink Average Throughput (DU)** before and after this function is enabled to monitor the benefits provided by this function.

5 NR Multi-Band Power Adaptive Scheduling (Low-Frequency TDD)

5.1 Principles

NR multi-band power adaptive scheduling implements power adaptive scheduling at the TTI level in a single RF module. [Figure 5-1](#) shows how this function works.

Figure 5-1 Process of NR multi-band power adaptive scheduling



This function is controlled by the **NR_MULTI_BAND_PWR_ADAPT_SCH_SW** option of the **NRDUCellAlgoSwitch.DynPowerSharingSwitch** parameter.

1. Setup of power adaptive scheduling groups

After this function is enabled, the gNodeB automatically sets up groups for power adaptive scheduling from NR cells served by an RF module when all of the following conditions are met:

- The RF module meets the requirement listed in [RF Modules](#).
- The cells in the same band have the same number of downlink transmit channels.
- The cells in the same band share PAs.
- The cells in the same band have the same slot configuration.
- The cells in the same band have the same subcarrier spacing.

Each BBP can accommodate a maximum of 12 power adaptive scheduling groups. In each group, at most two cells in the same NR band can participate in power adaptive scheduling.

2. Evaluation of the power adaptive scheduling status of each cell

The gNodeB checks the power adaptive scheduling status of each cell in a power adaptive scheduling group based on the volume of the data scheduled in the previous TTI. It evaluates whether a cell is receiving allocated power, allocating its power, or not participating in power adaptive scheduling.

3. Calculation of the power used by a cell

NR multi-band power adaptive scheduling cannot increase the RF channel capability, but it can increase the user-perceived throughput by reallocating power from idle cells to busy cells. When this function is enabled, the sum of power used by all cells in a power adaptive scheduling group cannot exceed the sum of their static power configurations, nor can it exceed the maximum rated power of their serving RF channel.

- For a cell receiving allocated power:

Power used by the cell in a TTI = Min{(Statically configured power + Received allocated power), Maximum rated power of the RF channel}

Specifically:

- Maximum allocated power that can be received by the cell = Max{(Statically configured power x Maximum percentage of allocated power), Maximum rated power of the RF channel}
- The maximum percentage of allocated power is specified by the **NRDUCellTrpDynPwrShr.MaxPowerSharingRate** parameter.
- The priority of a carrier to be involved in power adaptive scheduling is specified by the **NRDUCellTrpDynPwrShr.MultiBandPwrAdaptSchPri** parameter. A smaller value of this parameter indicates a lower priority in a power adaptive scheduling group, while a larger value indicates a higher priority.

 NOTE

"Statically configured power" refers to the power manually configured for a cell.
The maximum rated power of an RF channel is specific to the operating band of the local cell.

- For a cell allocating its power: Power used by the cell in a TTI = Statically configured power – Allocated power
- For a cell not participating in power scheduling: Power used by the cell in a TTI = Statically configured power

5.2 Network Analysis

5.2.1 Benefits

It is recommended that this function be enabled when traffic is instantaneously imbalanced between inter-band NR cells that do not share a channel of an RF module in light-load scenarios. This function increases the **User Downlink Average Throughput (DU)** by:

- **5% to 10%** if the percentage of the last TTIs before the downlink buffer is empty is less than 40%.
- **No more than 5%** if the percentage of the last TTIs before the downlink buffer is empty is greater than or equal to 40%.

 NOTE

Percentage of the last TTIs before the downlink buffer is empty (%) = $(N.ThpTime.DL - N.ThpTime.DL.RmvLastSlot) / N.ThpTime.DL \times 100\%$

The preceding benefits are obtained if both of the following conditions are met:

- The average downlink PRB usage is less than or equal to 40%.
Average downlink PRB usage (%) = $N.PRB.DL.Used.Avg / N.PRB.DL.Avail.Avg \times 100\%$
- The average downlink MCS index is less than or equal to 15.

 NOTE

UEs using 256QAM in the downlink do not benefit from NR multi-band power adaptive scheduling.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

- Changes in the application status of this function will trigger parameter adjustments, which cause traffic fluctuations for around 2 minutes and affect the cell throughput. Therefore, it is recommended that this function be enabled when traffic is light.
- Parameter adjustments following changes in the application status of this function change the coverage of SSB, CSI-RS, PDCCH, and control messages. Such changes may cause indicators related to the access success rate,

handover success rate, service drop rate, and paging success rate to deteriorate.

- This function increases the power consumed by the PDSCH, increasing the average MCS index and reducing the downlink PRB usage.
- The impact of this function on the downlink packet delay (indicated by the **N.QoS.DL.PktDelaygNBDU.Time** counter) is related to services. That is, when the percentage of tail packets (the percentage of the last TTIs before the downlink buffer is empty) is low, the downlink packet delay decreases. Conversely, when the percentage of tail packets is high, the downlink packet delay increases.
- This function increases the RF module power usage. Although a base station consumes more power, its power consumption does not exceed the maximum power consumption of all its cells working under full load.
- This function adjusts the power of the cells in a power adaptive scheduling group in each TTI, which requires information to be exchanged between the cells. This may slightly increase the CPU load of BBPs (indicated by **VS.BBUBoard.CPULoad.Mean**) to no more than 5%.
- This function evaluates the power adaptive scheduling status and allocates power at the TTI level based on scheduling results in the previous TTI. The evaluation may be inaccurate as NR services are running in real time. Therefore, the access delay increases. You are not advised to enable this function when testing the uplink and downlink peak data rates on the gNodeB.

5.3 Requirements

5.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
TDD	FOFD-090100	Multi-band Power Adaptive Scheduling	NR0S00MB PA00	gNode B	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Multi-position PDCCH scheduling	ANY_OCCASION_PD_CCH_SW option of the NRDUCellAlgoSwitch . LowLatencySwitch parameter	None
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch . DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch . DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch . DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>
Low-frequency TDD	NR intra-sector power sharing	NR_INTRA_SECTOR_POWER_SHARE_SW option of the NRDUCellAlgoSwitch . DynPowerSharingSwitch parameter	None
Low-frequency TDD	Slot-level power control	POWER_CONTROL_SLOT_SW option of the NRDUCellSmartPwrLock . SmartPowerLockSw parameter	None
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Antenna Config Mode and RRU Number ^a	SECTOREQM.ANTCFGMODE and SECTOREQM.RRUNUM	None
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE , and NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>
a: If the SECTOREQM.ANTCFGMODE parameter is set to BEAM and the SECTOREQM.RRUNUM parameter is set to 2 for the sector equipment identified by NRDUCellCoverage.SectorEqmId , NR multi-band power adaptive scheduling cannot be enabled.			

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power aggregation in unoccupied bandwidths	NRDUCellChnPwr.MaxPdschConvPwrOffset	<i>Power Control</i>	When NR multi-band power adaptive scheduling and power aggregation in unoccupied bandwidths are both enabled, the latter function may provide lower gains.
Low-frequency TDD	Power reduction for CCUs	NRDUCellChnPwr.MaxUeSatPwrOffset	<i>Power Control</i>	When NR multi-band power adaptive scheduling and power reduction for CCUs are both enabled, the former function may provide lower gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INTRF_AVOID_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	When interference coordination between public and dedicated networks and NR multi-band power adaptive scheduling are both enabled, the latter function may provide lower gains.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When intra-base-station DL CoMP and NR multi-band power adaptive scheduling are both enabled, the latter function may provide lower gains.
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW or INTER_GNB_DL_CS_CBF_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	When coordinated interference management and NR multi-band power adaptive scheduling are both enabled, the latter function may provide lower gains.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When Co-MM and NR multi-band power adaptive scheduling are both enabled, the latter function may provide lower gains.
Low-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When data convergence in slots is enabled for power saving scheduling in a cell, the cell can allocate less power to others, affecting the gains provided by NR multi-band power adaptive scheduling.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When NR multi-band power adaptive scheduling and energy saving based on flexible frequency-domain scheduling both take effect, both functions may provide lower gains.
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	A carrier exits a power adaptive scheduling group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	A carrier exits a power adaptive scheduling group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Cells in the energy saving state do not participate in power adaptive scheduling. When a cell enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: - In intra-base-station multi-carrier scenario s: INTRA_GNB_MULTICARRIER_SHUTDOWN option of the NRDualCellAlgorithmSwitch.PowerSavingSwitch parameter and FAST_CARRIER_SHUTDOWN option of the NRDualCellAlgorithmSwitch.PowerSavingExtensionSwitch parameter - In inter-base-station multi-carrier scenario s: INTER_GNB_M	<i>Energy Conservation and Emission Reduction</i>	A carrier exits a power adaptive scheduling group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
		ULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingSwitch</i> parameter and FAST_CARRIER_SHUT_DOWN_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingExtSwitch</i> parameter Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch. <i>PowerSavingExtSwitch</i> parameter		

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	NR multi-band power adaptive scheduling is not performed when dynamic RF channel shutdown is in effect.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCY_SW set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	A carrier exits a power adaptive scheduling group once it enters the deep dormancy state.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCY_SW (5G gNodeB, LTE eNodeB) set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	A carrier exits a power adaptive scheduling group once it enters the super dormancy state.
Low-frequency TDD	Non-Slot symbol-level scheduling for low latency	NON_SLOT_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter	None	The user-perceived rates of Non-Slot UEs may decrease in a cell allocating its power.
Low-frequency TDD	DL 256QAM	NRDUCellAlgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	UEs for which DL 256QAM is being used cannot benefit from NR multi-band power adaptive scheduling.
Low-frequency TDD	CSI-RS rate matching	CSIRS_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchingSwitch parameter	<i>Downlink Scheduling</i>	When CSI-RS rate matching takes effect, NR multi-band power adaptive scheduling may provide lower gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	NR multi-band power adaptive scheduling may provide lower gains in Hyper Cell networking.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUT DOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When NR multi-band power adaptive scheduling and TTI-level channel shutdown both take effect, the energy saving gains of TTI-level channel shutdown decrease and the downlink UE throughput may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency convergent scheduling	MULTI_FREQ_CONVERGENT_SCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	- Multi-frequency convergent scheduling enables real-time scheduling for individual CA UEs to solve the problem of inaccurate downlink data split between the PCell and SCells. - NR multi-band power adaptive scheduling allocates the estimated remaining power of a carrier to other carriers for a CA UE if the carrier is not allocated enough data as a result of inaccurate downlink data split between the PCell and SCells. Therefore, both multi-frequency convergent scheduling and NR multi-band power adaptive scheduling can improve the downlink experience of CA UEs. When both functions are enabled, the benefits provided by either function may decrease.
Low-frequency TDD	PDCCH aggregation level reduction	PDCCH_AGG_LVL_COMPRESS_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Only NR multi-band power adaptive scheduling takes effect when it is enabled together with PDCCH aggregation level reduction.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellDMAc.DlModulationSchemeSw parameter	<i>Modulation Schemes</i>	If there are UEs for which DL 1024QAM is being used, NR multi-band power adaptive scheduling provides lower gains.
Low-frequency TDD	Low power consumption mode (NR)	LOW_PWR_MODE_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	When both low power consumption mode (NR) and NR multi-band power adaptive scheduling are enabled and a customized backup power saving policy is used, only low power consumption mode takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Maximum SSB power offset	NRDUCellTrpBeam.MaxSsbPwrOffset	<i>Power Control</i>	When the NRDUCellTrpBeam.MaxSsbPwrOffset parameter is set to a value greater than 3 dB and NR multi-band power adaptive scheduling is in effect: • If the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter is deselected, the SSB power aggregation capability decreases. • If the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter is selected, NR inter-carrier dynamic power sharing does not work in slots with SSB transmission. This prevents the decrease in the SSB power aggregation capability, but this power sharing function no longer delivers gains in these slots.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell load simulation	STRNRDUCELLSIMULATELOAD	None	If both cell load simulation and multi-band power adaptive scheduling take effect, it is recommended that the CSIRS_ACC_PRED_SW option of the NRDUCellServExp.ServicExpAlgoSwitch parameter be selected to increase the gains of multi-band power adaptive scheduling in load simulation scenarios.
Low-frequency TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When NR multi-band power adaptive scheduling takes effect, CCE partial symbol adaptation provides lower gains.
Low-frequency TDD	Downlink precise RB estimation	PRECISE_DL_RB_ESTIMATE_SW option of the NRDUCellDISch.DISchAlgoSwitch parameter	<i>Downlink Scheduling</i>	When downlink precise RB estimation and NR multi-band power adaptive scheduling are both enabled, the latter function may provide lower gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	A carrier exits a power sharing group once RF module fast deep dormancy takes effect for this carrier. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

The main control board must be a UMPT board, and the baseband processing unit must be a UBBPfw, UBBPg, or NR TDD-capable board of a later model.

RF Modules

AAU5368md (whose SBOM is not 02315DBT) and AAU5665md

5.3.4 Cells

- **NRDUCell.DuplexMode** must be set to **CELL_TDD**.
- **NRDUCellTrp.TxRxMode** must be set to **64T64R** or **32T32R**.

- Among cells served by the same RF module, NR multi-band power adaptive scheduling can be enabled for only two of them in each NR band.

5.3.5 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-1](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dynamic Power Sharing Switch	NRDUCellAlgoSwitch. <i>DynPowerSharingSwitch</i>	Select the NR_MULTI_BAND_PWR_ADAPT_SCH_SW option if NR multi-band power adaptive scheduling is required.
Max Power Sharing Rate	NRDUCellTrpDynPwrShr. <i>MaxPowerSharingRate</i>	Use the recommended value.
Multi-Band Power Adapt Scheduling Priority	NRDUCellTrpDynPwrShr. <i>MultiBandPwrAdaptSchPri</i>	Use the recommended value.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for NR multi-band power adaptive scheduling for the cell whose ID is 0
MOD NRDUCELLALGOSWITCH:
```

```
NrDuCellId=0,DynPowerSharingSwitch=NR_MULTI_BAND_PWR_ADAPT_SCH_SW-1;  
//Setting the MaxPowerSharingRate parameter and priority of multi-band power adaptive scheduling for  
the NR DU cell TRP whose ID is 0  
MOD NRDUCELLTRPDYNPWRSHR: NrDuCellTrpId=0, MaxPowerSharingRate=100,  
MultiBandPwrAdaptSchPri=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for NR multi-band power adaptive scheduling for the cell whose ID is 0  
MOD NRDUCELLALGOSWITCH:  
NrDuCellId=0,DynPowerSharingSwitch=NR_MULTI_BAND_PWR_ADAPT_SCH_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

To verify that this function has been activated, perform the following steps:

- Step 1** Run the **DSP NRDUCELLTRPDYNPWRSHR** command on the gNodeB. If the value of **NR Multi-Band Adapt Pwr Sch Valid Status** is **Enabled**, this function has been enabled.
- Step 2** Run the **DSP GNODEBDYNPWRSHRGRP** command on the gNodeB to check whether power adaptive scheduling groups are successfully set up and which cells are in the same group.
- Step 3** Observe performance counters. If neither of the counters listed in [Table 5-2](#) is zero, this function has taken effect.

Table 5-2 Counters related to NR multi-band power adaptive scheduling

Counter ID	Counter Name
1911823028	N.PwrShare.PwrOut.Slot.Num
1911823027	N.PwrShare.PwrIn.Slot.Num

NOTE

The **N.PwrShare.PwrIn.Slot.Num** counter indicates the actual number of TTIs within which the cell uses received allocated power. As NR multi-band power adaptive scheduling uses the estimation results from the previous TTI, the actual number of TTIs within which the cell uses the received power might differ from the estimated number.

----End

5.4.3 Network Monitoring

Compare the values of **User Downlink Average Throughput (DU)** before and after this function is enabled to monitor the benefits provided by this function.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- *CoMP*
- *Energy Conservation and Emission Reduction*
- *Multi-RAT Coordinated Energy Saving*
- *Cell Combination*
- *LTE and NR Power Allocation*
- *Modulation Schemes*
- *High Speed Mobility*
- *Co-MM (Low-Frequency TDD)*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *Power Control*
- *Downlink Scheduling*
- *Multi-Frequency Smart Aggregation*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *License Control Item Lists*
- *Green Site*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Hyper Cell*
- *Fusion Cell (Low-Frequency TDD)*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"